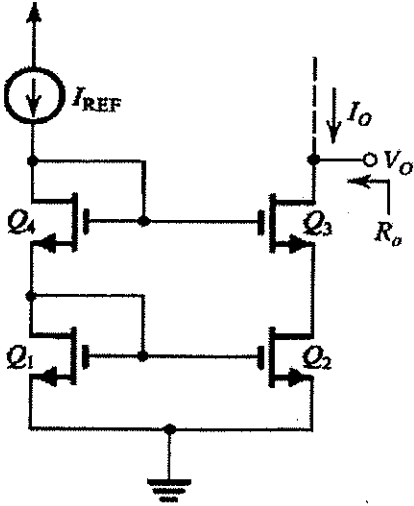


B. Tech (ECE) Semester: III
MID-SEMESTER EXAMINATION, September 2023

Max. Marks: 15
 Duration: 1:30 Hours

Course Title: **Microelectronics**
 Course Code: ECECC07

Note: - Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

Q.No.	Question	Marks	CO
1a	Draw the high-frequency model of a MOSFET and explain the key parameters of the model and their significance. What is the expression for the Unity-Gain Frequency (f_T)?	2	1
1b	What is meant by the <i>compliance voltage</i> of a current mirror? Compare the compliance voltages of the following BJT mirrors: the simple Current mirror, Wilson Current Mirror and Cascode current mirror.	1	1
2a	For the circuit shown in Fig. 1, all transistors have $V_{th}=0.6V$, $\mu_n C_{ox}=160\mu A/V^2$, $L=1\mu m$, $V_A=10V$, Widths $W_1=W_4=4\mu m$ and $W_2=W_3=40\mu m$ and $I_{REF}=20\mu A$. Determine R_o and the lowest voltage at the output for which current source operation is possible.  Fig. 1	2	1
2b	Derive an expression for the small signal output resistance of the BJT Wilson Current Mirror.	1	1
3a	Perform the high-frequency analysis of a common emitter amplifier and derive the expression for its: (i) voltage gain and (ii) output resistance.	2	1
3b	Find the mid-band gain A_M and the upper 3-dB frequency f_H of a CS amplifier for the circuit shown in Fig. 2. The amplifier has $R_{sig}=100k\Omega$, $R_G=4.7M\Omega$, $R_D=R_L=15k\Omega$, $g_m=1mA/V$, $r_o=150k\Omega$, $C_{gs}=1pF$, and $C_{gd}=0.4pF$.	1	1

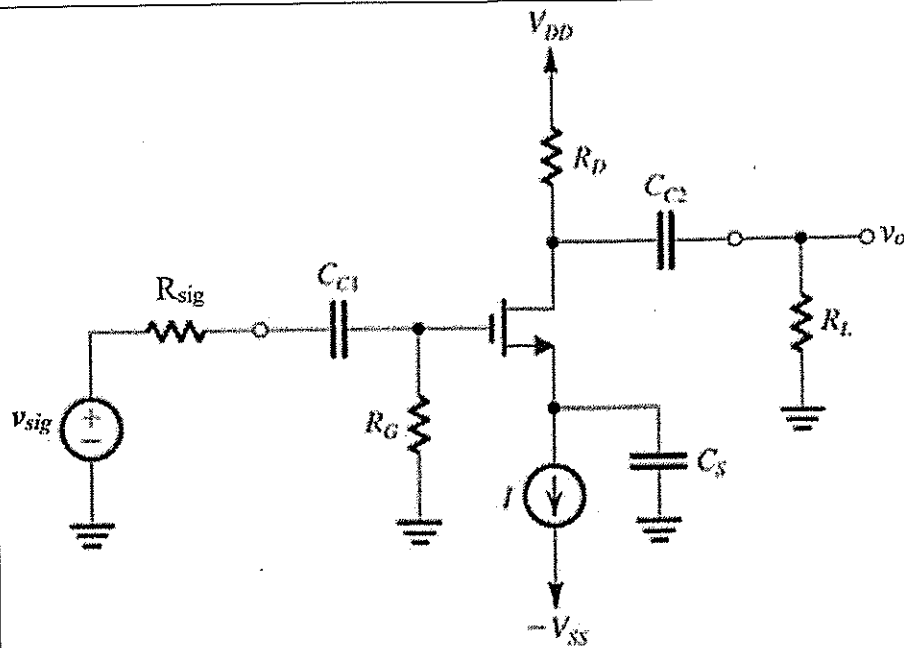


Fig. 2

4a	By appropriate analysis, determine the linear operational range of input differential signals for an NMOS differential amplifier.	2	2
4b	Explain the concept of <i>active load</i> as applicable to a common source amplifier.	1	2
5a	Explain the small signal operation of bipolar differential pair for differential input voltage and derive an expression for its <i>trans-conductance</i> G_m .	2	2
5b	Explain Miller's Theorem and its importance in the analysis of transistor amplifiers.	1	2